

Question

Graphdiyne is a novel carbon material. Similar to graphene, graphdiyne consists of a single-layer network structure composed of sp^2 and sp hybridized carbon, with layers connected by van der Waals forces. Because sp^2 hybridized carbon and sp hybridized carbon can have various arrangements, graphdiyne can have multiple structures. In one type of graphdiyne, each $C(sp^2)$ is connected to three $C(sp)$, and each $C(sp)$ is connected to one $C(sp)$ and one $C(sp^2)$ respectively, which we temporarily name graphdiyne *A* here; in another type of graphdiyne, each $C(sp^2)$ is connected to two $C(sp)$ and one $C(sp^2)$ respectively, and each $C(sp)$ is connected to one $C(sp)$ and one $C(sp^2)$ respectively, which we temporarily name graphdiyne *B* here; in addition, there is a graphdiyne structure consisting of a network structure composed of $[C(sp^2)C(sp)C(sp)C(sp^2)]$, but containing a benzene ring structure, which we temporarily name graphdiyne *C* here.

The graphdiyne *C* variant, graphdienes, can be synthesized on any substrate using hexaethynylbenzene as a monomer. The method involves wrapping the substrate material in a metal mesh and immersing it in an acetone solution containing hexaethynylbenzene in pyridine and tetramethylethylenediamine, which can induce a Glaser-Hay alkyne coupling reaction on the substrate to obtain the grown graphdienes.

Based on the above background information, which of the following statements is incorrect ()?

- A. Graphdiyne *A* has six sp hybridized carbons in the primitive cell.
- B. Metal mesh elements have two stable ionic valence states.
- C. Graphdiyne *C* can be synthesized from monomers containing three bromine atoms.
- D. Graphdiyne *A* can be synthesized from a monomer containing two bromine atoms, and this monomer cannot have cis-trans isomers.
- E. Graphene-yne *A* and graphene-yne *C* have a ratio of 2:3 in the number of carbon atoms contained in their primitive unit cells.

- F.** Graphdiyne C adopts an ABAB layered stacking (alternating concentric large and small rings), and after sodium ions are intercalated into the interlayer, it forms a NaC composition (i.e., Na:C=1:1). If the polygons in its top view represent carbon skeleton pores (carbon atoms are not drawn), and the dots represent sodium ions, then the sodium ions are located at the center of the regular hexagon formed by the overlapping projection of layer A and layer B, with one sodium ion per hexagon.

Answer

Correct Answer: **F**

Detailed Explanation

A: Graphdiyne A is an 18-membered ring arrangement of hexagons composed of $[C(sp^2)C(sp)C(sp)C(sp^2)]$ cycles. According to the definition and division principles of a proper unit cell, its proper unit cell is a rhombus containing 8 carbon atoms [a rhombus formed by connecting the centers of four 18-membered rings in adjacent rows (taking two 18-membered rings each from the upper and lower rows)], which contains 6 sp hybridized carbons and 2 sp^2 hybridized carbons.

CHECKPOINT

2 PTS

The proper unit cell of graphdiyne A is a rhombus containing 8 carbon atoms [a rhombus formed by connecting the centers of four 18-membered rings in adjacent rows (taking two 18-membered rings each from the upper and lower rows)]

B: The copper-catalyzed cross-coupling reaction of terminal alkynes (Glaser-Hay reaction) is a fundamental reaction in organic chemistry. It is inferred that the metal mesh is a copper mesh, and $Cu(py)_2^+$ can migrate to the substrate for adsorption, and then the catalytic coupling reaction occurs on the substrate surface. The stable valence states of copper ions include +1 and +2.

CHECKPOINT

2 PTS

Copper-catalyzed cross-coupling reaction of terminal alkynes (Glaser-Hay reaction)

C: Graphdiyne *C* can be synthesized from the monomer C#CC1=C(Br)C(C#C)=C(Br)C(C#C)=C1Br, which contains 3 bromine atoms.

CHECKPOINT

1 PTS

Graphdiyne *C* can be synthesized from the monomer C#CC1=C(Br)C(C#C)=C(Br)C(C#C)=C1Br

D: Graphdiyne *A* can be synthesized from the Br/C(Br)=C(C#C)\C#C monomer and has no cis-trans isomers.

CHECKPOINT

1 PTS

Graphdiyne *A* can be synthesized from the Br/C(Br)=C(C#C)\C#C monomer and has no cis-trans isomers

E. The proper unit cell of graphdiyne *C* contains a benzene ring and six **sp** hybridized carbons, a total of 12 carbon atoms. As described in the analysis of option A, the proper unit cell of graphdiyne *A* contains six **sp** hybridized carbons and two **sp²** hybridized carbons, a total of 8 carbon atoms.

CHECKPOINT

1 PTS

The proper unit cell of graphdiyne *C* contains a benzene ring and six **sp** hybridized carbons. The proper unit cell of graphdiyne *A* contains six **sp** hybridized carbons and two **sp²** hybridized carbons.

F. ABAB stacking means that the rings of adjacent layers (A and B) are not aligned, but the cocentric rings of different sizes are staggered, so the polygon projections of layers A and B should be staggered in the top view. The description "sodium ions are filled in the interlayer and the position of sodium ions in each layer is the same" indicates that sodium ions are located between the two layers. If the sodium ions are located in the center of the overlapping area, the overlapping area may not be centered after the AB layers are misaligned. "Sodium ions are

located in the center of the regular hexagon formed by the overlapping projections of layers A and B" is incorrect because the misalignment of AB stacking causes the overlapping area of the projection to no longer be the center of the original hexagon. Therefore, the interlayer sodium ions should occupy the projection of the intersection of three adjacent polygons, similar to the intercalation sites of lithium ions in graphite. Sodium ions are located at the trifurcated voids where the vertices of adjacent A and B layer polygons intersect, with one sodium ion at each trifurcation point.

CHECKPOINT

2 PTS

If the sodium ions are located in the center of the overlapping area, the overlapping area may not be centered after the AB layers are misaligned.